

scaling centralizes all transaction processing to a single large machine. This centralization reduces the communication overhead (transaction/block latency) associated with a PoW blockchain such as Ethereum but results in a centralized architecture in which one machine is responsible for most of the system's processing. Some blockchains, such as Solana,<sup>17</sup> follow this approach and can achieve upward of 50,000 TPS.

Horizontal scaling, however, divides the work of the system into multiple pieces, retaining decentralization but increasing the throughput of the system through parallelization. *Ethereum 2.0* takes this approach (called *sharding*) in combination with a proof-of-stake consensus algorithm.

Ethereum 2.0's technical architecture<sup>18</sup> differs drastically from vertically scaled blockchains such as Solana, but the improvements are the same. Ethereum 2.0 uses horizontal scaling with multiple blockchains and can achieve upward of 50,000 TPS.

The development of Ethereum 2.0 has been delayed for several years, but its mainnet, which will contain a basic blockchain without any smart contract support, may go live in 2021. Ethereum 2.0 has not yet finalized a functional specification for sending transactions between its horizontally scaled blockchains.

Another idea with the potential to reduce scaling risk is the Ethereum layer-2 landscape. *Layer 2* refers to a solution built on top of a blockchain that relies on cryptography and economic guarantees to maintain desired levels of security.